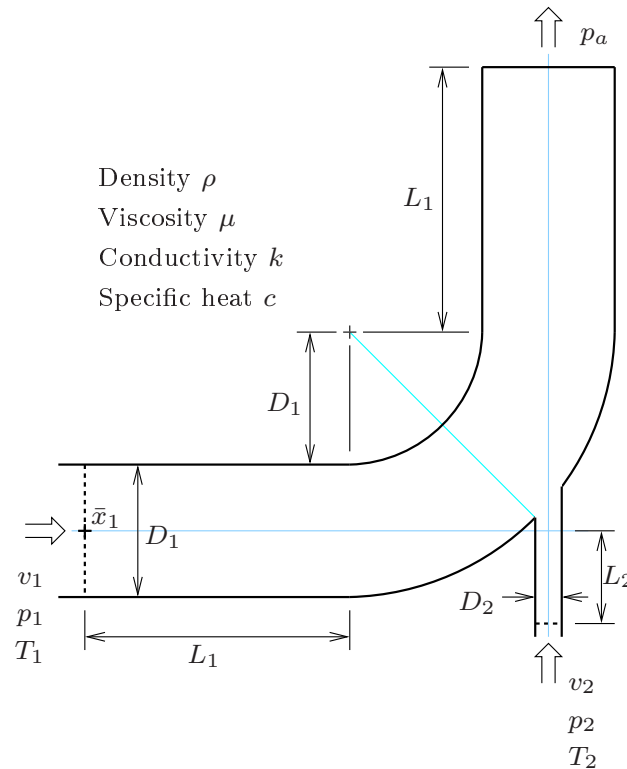


PROBLEM 1 (50 minutes)

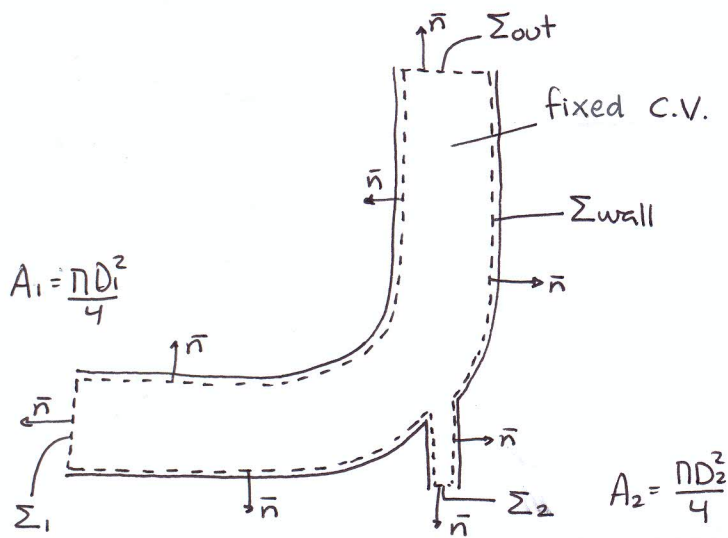
Consider the mixing elbow shown schematically in the figure. Cold water flows through the main pipe and mixes with warmer water that enters through a smaller inlet located at the elbow. The liquid properties can be considered independent of the temperature and gravity effects can be neglected. All pipe walls are adiabatic. The known pipe dimensions, fluid properties and boundary conditions are given in the figure.



Assume that the two streams mix completely and that the velocity and temperature are uniform at the exit section.

1. (3 points). Find the velocity v_s and temperature T_s of the mixed stream emerging from the pipe at ambient pressure.
2. (3 points). Obtain the net force $\bar{F}$ exerted on the mixing elbow by the water flowing inside and the ambient air.
3. (3 points). Obtain the torque $\bar{M}_1$ about the point $\bar{x}_1$ located at the center of the larger inlet.
4. (1 point). Compute the above results numerically, given the following data:

$v_1 = 0,4 \text{ m/s};$	$v_2 = 1,2 \text{ m/s};$	$\rho = 1000 \text{ kg/m}^3;$
$T_1 = 20^\circ\text{C};$	$T_2 = 40^\circ\text{C};$	$\mu = 8 \cdot 10^{-4} \text{ Pa s};$
$p_1 - p_a = 50 \text{ Pa};$	$p_2 - p_a = 200 \text{ Pa};$	$k = 0,677 \text{ W/(m s)};$
$D_1 = 10 \text{ cm};$	$D_2 = 2,5 \text{ cm};$	$c = 4216 \text{ J/(kg K)};$
$L_1 = 20 \text{ cm};$	$L_2 = 7 \text{ cm}.$	



④

$$v_s = 0.475 \frac{m}{s}$$

$$T_s = 23^\circ C$$

$$\vec{F} = 1.65 N \vec{e}_x - 0.967 N \vec{e}_y$$

$$\vec{M}_1 = -0.338 Nm \vec{e}_z$$

① MASS CONSERVATION : $v_1 A_1 + v_2 A_2 = v_s A_s \rightarrow$

$$v_s = v_1 + v_2 \left(\frac{D_2}{D_1} \right)^2$$

ENERGY CONSERVATION:

$$-\rho \left(e_1 + \frac{p_1}{\rho} + \frac{v_1^2}{2} \right) v_1 A_1 - \rho \left(e_2 + \frac{p_2}{\rho} + \frac{v_2^2}{2} \right) v_2 A_2 + \rho \left(e_s + \frac{p_a}{\rho} + \frac{v_s^2}{2} \right) v_s A_s = 0$$

$$e = e_0 + cT$$

$$T_s = \left(T_1 + \frac{p_1 - p_a}{\rho c} + \frac{v_1^2}{2c} \right) \frac{v_1}{v_s} + \left(T_2 + \frac{p_2 - p_a}{\rho c} + \frac{v_2^2}{2c} \right) \frac{v_2 A_2}{v_s A_s} - \frac{v_s^2}{2c}$$

② FORCE $\vec{F} = - \int_{\Sigma_{wall}} [-(p - p_a) \vec{n} + \vec{\bar{z}}' \cdot \vec{n}] d\sigma$

MOMENTUM CONSERVATION:

$$-\rho v_1^2 A_1 \vec{e}_x - \rho v_2^2 A_2 \vec{e}_y + \rho v_s^2 A_s \vec{e}_y = (p_1 - p_a) A_1 \vec{e}_x + (p_2 - p_a) A_2 \vec{e}_y - \vec{F}$$

$$\vec{F} = (\rho v_1^2 + p_1 - p_a) A_1 \vec{e}_x + [(\rho v_2^2 + p_2 - p_a) A_2 - \rho v_s^2 A_s] \vec{e}_y$$

③ TORQUE $\vec{M}_1 = - \int_{\Sigma_{wall}} (\vec{x} - \vec{x}_1) \wedge [-(p - p_a) \vec{n} + \vec{\bar{z}}' \cdot \vec{n}] d\sigma$

ANGULAR MOMENTUM CONSERVATION:

$$-\rho v_2^2 A_2 (\vec{x}_2 - \vec{x}_1) \wedge \vec{e}_y + \rho v_s^2 A_s (\vec{x}_s - \vec{x}_1) \wedge \vec{e}_y = (p_2 - p_a) A_2 (\vec{x}_2 - \vec{x}_1) \wedge \vec{e}_y - \vec{M}_1$$

$$\vec{M}_1 = [(\rho v_2^2 + p_2 - p_a) A_2 - \rho v_s^2 A_s] \left(L_1 + \frac{3}{2} D_1 \right) \vec{e}_z$$